



Technical Report

Building and Securing a Mini Production-Ready Web Infrastructure

Course: Web Infrastructure

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Technical Report: Building and Securing a Mini Production-Ready Web Infrastructure

1. Introduction

This report documents the setup and configuration of a mini production-ready web infrastructure using Windows Subsystem for Linux (WSL) running Ubuntu. The project demonstrates core infrastructure concepts, including remote access, web server configuration, API interaction, load balancing, security implementation, and system debugging.

The work simulates the preparation tasks performed by our IT infrastructure team before deploying a live application.

2. Environment and Tools Used

- ♦ **Operating System:** Windows11 with WSL
- ♦ **Linux Distribution:** Ubuntu (WSL)
- ♦ **Web Server:** Nginx
- ♦ **SSH:** OpenSSH
- ♦ **Firewall:** UFW
- ♦ **SSL/TLS:** OpenSSL
- ♦ **API Testing:** Postman

```
rutaganira@DESKTOP-BB7D3I x + v
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>wsl
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$
```

3. Remote Server Simulation (SSH)

SSH was configured on the local WSL instance to enable secure remote access to the simulated server. The system was updated before installing OpenSSH server.

```
sudo apt update
```

```
sudo apt upgrade
```

```
Sudo apt install openssh-server
```

3.1 Password-Based Authentication

After installation, the SSH service was started and verified to be running.

```
rutaganira@DESKTOP-BB7D3I x + v
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$ sudo service ssh start
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Sat 2026-01-10 13:52:45 +04; 1min 8s ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 7435 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 7436 (sshd)
    Tasks: 1 (limit: 9287)
   Memory: 1.4M (peak: 1.8M)
      CPU: 16ms
   CGroup: /system.slice/ssh.service
           └─7436 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 10 13:52:45 DESKTOP-BB7D3DM systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 10 13:52:45 DESKTOP-BB7D3DM sshd[7436]: Server listening on 0.0.0.0 port 22.
Jan 10 13:52:45 DESKTOP-BB7D3DM sshd[7436]: Server listening on :: port 22.
Jan 10 13:52:45 DESKTOP-BB7D3DM systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$ |
```

The SSH connection was tested from another terminal using password authentication. Basic commands were run to verify successful connection:

```

PS C:\Users\USER> ssh rutaganira@172.25.17.49
The authenticity of host '172.25.17.49 (172.25.17.49)' can't be established.
ED25519 key fingerprint is SHA256:zSPVdqQqHoDuUkMJaa/4TJQT3vC8I1D5y8FZUdFLPso.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.25.17.49' (ED25519) to the list of known hosts.
ssh_dispatch_run_fatal: Connection to 172.25.17.49 port 22: Connection aborted
PS C:\Users\USER> ssh rutaganira@172.25.17.49
rutaganira@172.25.17.49's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 10 14:06:50 +04 2026

System load:  0.0          Processes:           35
Usage of /:   0.2% of 1006.85GB  Users logged in:    1
Memory usage: 5%          IPv4 address for eth0: 172.25.17.49
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 10 14:06:50 +04 2026

System load:  0.0          Processes:           35
Usage of /:   0.2% of 1006.85GB  Users logged in:    1
Memory usage: 5%          IPv4 address for eth0: 172.25.17.49
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

```

3.2 Key-Based Authentication

An SSH key pair was generated using `ssh-keygen`. The public key was added to the server's `authorized keys` file, enabling password less authentication with improved security.

After configuration, SSH login was tested using the private key instead of password authentication.

```

rutaganira@DESKTOP-BB7D3C  X  rutaganira@DESKTOP-BB7D3C  X  +  v
PS C:\Users\USER> ssh -i $env:USERPROFILE\.ssh\wsl_key rutaganira@172.25.17.49
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 10 14:51:12 +04 2026

System load:  0.0          Processes:           33
Usage of /:   0.2% of 1006.85GB  Users logged in:    1
Memory usage: 5%          IPv4 address for eth0: 172.25.17.49
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sat Jan 10 14:31:37 2026 from 172.25.16.1
rutaganira@DESKTOP-BB7D3DM:~$ |

```

3.3 File Transfer

Secure file transfer was performed using SCP (Secure Copy Protocol) over the SSH connection.

```
rutaganira@DESKTOP-BB7D3C x rutaganira@DESKTOP-BB7D3C x Windows PowerShell x + v
PS C:\Users\USER> scp -i $env:USERPROFILE\.ssh\authorized_key testfile.txt rutaganira@172.25.17.49:/home/
Warning: Identity file C:/Users/USER/.ssh/authorized_key not accessible: No such file or directory.
rutaganira@172.25.17.49's password:
Connection closed by 172.25.17.49 port 22
C:\Windows\System32\OpenSSH\scp.exe: Connection closed
PS C:\Users\USER> scp -i $env:USERPROFILE\.ssh\wsl_key testfile.txt rutaganira@172.25.17.49:/home
C:\Windows\System32\OpenSSH\scp.exe: dest open "/home/testfile.txt": Permission denied
C:\Windows\System32\OpenSSH\scp.exe: failed to upload file testfile.txt to /home
PS C:\Users\USER> scp -i $env:USERPROFILE\.ssh\wsl_key testfile.txt rutaganira@172.25.17.49:~/
testfile.txt
100% 44 43.0KB/s 00:00
PS C:\Users\USER> echo "Hello from WSL!" > wsl_file.txt
PS C:\Users\USER> cat wsl
cat : Cannot find path 'C:\Users\USER\wsl' because it does not exist.
At line:1 char:1
+ cat wsl
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\USER\wsl:String) [Get-Content], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\Users\USER> cat wsl_file.txt
Hello from WSL!
PS C:\Users\USER>
```

4. Web Server and HTTP Protocol Demonstration

Nginx was installed and configured to serve content on the local system.

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 8 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 4s (142 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 43692 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.5) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
```

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx start
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx status
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-16 16:18:15 +04; 1min 7s ago
     Docs: man:nginx(8)
  Process: 1071 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 1072 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1110 (nginx)
      Tasks: 13 (limit: 9287)
    Memory: 9.0M (peak: 20.1M)
       CPU: 94ms
    CGroup: /system.slice/nginx.service
            └─1110 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─1113 "nginx: worker process"
                  └─1114 "nginx: worker process"
                     └─1115 "nginx: worker process"
                        └─1116 "nginx: worker process"
                           └─1117 "nginx: worker process"
                              └─1118 "nginx: worker process"
                                 └─1119 "nginx: worker process"
                                    └─1120 "nginx: worker process"
                                       └─1121 "nginx: worker process"
                                          └─1122 "nginx: worker process"
                                             └─1123 "nginx: worker process"
                                                └─1124 "nginx: worker process"

Jan 16 16:18:15 DESKTOP-BB7D3DM systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 16 16:18:15 DESKTOP-BB7D3DM systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

1.1 Custom Web Pages

Custom HTML pages were created to demonstrate basic web hosting functionality:

- Home page (index.html)

- About page (about.html)
- Contact page (contact.html)

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo nano /var/www/html/index.html
[sudo] password for rutaganira:
rutaganira@DESKTOP-BB7D3DM:~$ sudo nano /var/www/html/about.html
rutaganira@DESKTOP-BB7D3DM:~$ sudo nano /var/www/html/contact.html
rutaganira@DESKTOP-BB7D3DM:~$ sudo nano /var/www/html/about.html
rutaganira@DESKTOP-BB7D3DM:~$ sudo nano /var/www/html/about.html
rutaganira@DESKTOP-BB7D3DM:~$
```

4.1 HTTP Methods and Headers

The main HTTP methods (GET, POST, PUT, DELETE) were tested using curl commands to demonstrate how the web server processes different request types.

Test 1: Get

```
rutaganira@DESKTOP-BB7D3DM:~$ curl -v http://localhost
* Host localhost:80 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:80...
* Connected to localhost (::1) port 80
> GET / HTTP/1.1
> Host: localhost
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Server: nginx/1.24.0 (Ubuntu)
< Date: Fri, 16 Jan 2026 13:43:55 GMT
< Content-Type: text/html
< Content-Length: 4526
< Last-Modified: Fri, 16 Jan 2026 13:06:42 GMT
< Connection: keep-alive
< ETag: "696a37e2-11ae"
< Accept-Ranges: bytes
```

Test 2: GET with Custom Headers

```
rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$ curl -i -X GET http://172.25.17.49/api/contact
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Sun, 18 Jan 2026 21:18:33 GMT
Content-Type: text/plain
Content-Length: 19
Connection: keep-alive

GET /api/contact OKrutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$ |
```

Test 3: Post request

```
rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$ curl -i -X POST http://172.25.17.49/api/contact
HTTP/1.1 201 Created
Server: nginx/1.24.0 (Ubuntu)
Date: Sun, 18 Jan 2026 21:28:04 GMT
Content-Type: text/plain
Content-Length: 25
Connection: keep-alive

POST /api/contact Created rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$
```

Test 4: Put request

```
POST /api/contact Created rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$ curl -i -X PUT http://172.25.17.49/api/contact
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Sun, 18 Jan 2026 21:28:52 GMT
Content-Type: text/plain
Content-Length: 19
Connection: keep-alive

PUT /api/contact OK rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$ |
```

Test 5: Delete request

```
PUT /api/contact OK rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$ curl -i -X DELETE http://172.25.17.49/api/contact
HTTP/1.1 204 No Content
Server: nginx/1.24.0 (Ubuntu)
Date: Sun, 18 Jan 2026 21:29:23 GMT
Connection: keep-alive

rutaganira@DESKTOP-BB7D3DM:/etc/nginx/sites-available$
```


4.2 HTTP Status Codes summary

Various HTTP status codes were observed during testing, including 200 (OK), 301 (Redirect), 403 (Forbidden), and 204 (No content). Request and response headers were inspected using curl with verbose output.

Method	Expected Status Code	Meaning
GET	200 OK	Request successful
POST	201 Created	Resource created
PUT	200 OK	Resource updated
DELETE	204 No Content	Resource deleted

4.3 Log Files

Nginx access logs were examined to verify incoming requests and server responses.

Sudo tail -n 20/var/log/nginx/access.log

```
rutaganira@DESKTOP-BB7D3DM: /etc/nginx/sites-available$ sudo tail -n 20 /var/log/nginx/access.log
172.25.16.1 - - [19/Jan/2026:01:10:53 +0400] "GET /contact.html HTTP/1.1" 304 0 "http://172.25.17.49/contact.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36"
172.25.16.1 - - [19/Jan/2026:01:10:57 +0400] "GET /contact.html HTTP/1.1" 304 0 "http://172.25.17.49/contact.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36"
172.25.16.1 - - [19/Jan/2026:01:11:00 +0400] "GET / HTTP/1.1" 304 0 "http://172.25.17.49/contact.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36"
172.25.16.1 - - [19/Jan/2026:01:11:07 +0400] "GET /contact.html HTTP/1.1" 304 0 "http://172.25.17.49/about.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36"
172.25.16.1 - - [19/Jan/2026:01:11:44 +0400] "GET /contact.html HTTP/1.1" 304 0 "http://172.25.17.49/about.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36"
172.25.17.49 - - [19/Jan/2026:01:16:41 +0400] "GET /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.16.1 - - [19/Jan/2026:01:17:36 +0400] "GET /contact.html HTTP/1.1" 200 1293 "http://172.25.17.49/about.html" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36"
172.25.17.49 - - [19/Jan/2026:01:18:33 +0400] "GET /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:19:39 +0400] "POST /api/contact HTTP/1.1" 200 20 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:24:46 +0400] "GET /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:24:59 +0400] "POST /api/contact HTTP/1.1" 201 25 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:25:36 +0400] "PUT /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:26:56 +0400] "DELETE /api/contact HTTP/1.1" 204 0 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:27:26 +0400] "PUT /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:28:04 +0400] "POST /api/contact HTTP/1.1" 201 25 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:28:52 +0400] "PUT /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:29:23 +0400] "DELETE /api/contact HTTP/1.1" 204 0 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:33:33 +0400] "GET /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:38:15 +0400] "GET /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
172.25.17.49 - - [19/Jan/2026:01:40:16 +0400] "GET /api/contact HTTP/1.1" 200 19 "-" "curl/8.5.0"
rutaganira@DESKTOP-BB7D3DM: /etc/nginx/sites-available$
```

5. API Interaction

We used Json public API to demonstrate RESTful API communication using Postman and curl. A personal access token was generated for authenticated requests.

API Testing Tools: Postman, curl

API Endpoint: JSON REST API

Authentication: Personal Access Token (PAT)

5.1 POSTMAN COLLECTION RUNNER

API Test - Run results ▶ Run Again + New Run 📄 Automate Run Share ⋮

 Ran today at 12:01:10 AM · [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	5s 1ms	0	0	635 ms

[All Tests](#) [Passed \(0\)](#) [Failed \(0\)](#) [Skipped \(0\)](#) [Errors \(0\)](#) [Console Log](#) [View Summary](#)

POST Create a new post.
https://jsonplaceholder.typicode.com/posts

201 • 1553 ms • 1.327 KB •

No tests found

GET wrong API
https://jsonplaceholder.typicode.com/posts/99999

404 • 16 ms • 1.159 KB •

No tests found

PUT Update Post
https://jsonplaceholder.typicode.com/posts/1

200 • 1573 ms • 1.227 KB •

No tests found

DELETE delete a post
https://jsonplaceholder.typicode.com/posts/1

200 • 535 ms • 1.144 KB •

No tests found

GET Get All Users
https://jsonplaceholder.typicode.com/users

200 • 16 ms • 3.011 KB •

No tests found

5.2 Authentication and Error Handling

We use an invalid endpoints to observe API error responses and status codes.

GET | | Send

Docs Params **Authorization** Headers (7) Body Scripts Settings Cookies

Auth Type

Inherit auth from parent

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

[Edit Auth in collection](#)



No Auth

This request does not use any authorization. ⓘ

Body Cookies Headers (24) Test Results | 404 Not Found • 1.49 s • 1.12 KB • Save Response

{ } JSON | Preview | Debug with AI | 1 { }

GET | | Send

Docs Params Authorization Headers (9) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON | Schema Beautify

```
1 {
2   "title": "This should fail"
3 }
4
```

Response | ⌚



Could not get response

! Error: socket hang up | Debug with AI

View in Console • Learn about troubleshooting requests

5.3 USING CURL OPERATION ON THE TERMINAL

Test 1: GET /posts/1 (Retrieve a post)

```
rutaganira@DESKTOP-BB7D3D3: /mnt/c/Users/USER$ curl -s -X GET https://jsonplaceholder.typicode.com/posts/1 \
-w "\nStatus Code: %{http_code}\n"
{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut
eniet architecto"
}
Status Code: 200
rutaganira@DESKTOP-BB7D3D3: /mnt/c/Users/USER$
```

Test 2: POST /posts (Create a new post)

```
rutaganira@DESKTOP-BB7D3D3: /mnt/c/Users/USER$ curl -s -X POST https://jsonplaceholder.typicode.com/posts \
-H "Content-Type: application/json" \
-d '{
  "title": "My Test Post",
  "body": "This is a test post",
  "userId": 1
}' \
-w "\nStatus Code: %{http_code}\n"
{
  "title": "My Test Post",
  "body": "This is a test post",
  "userId": 1,
  "id": 101
}
Status Code: 201
rutaganira@DESKTOP-BB7D3D3: /mnt/c/Users/USER$
```

Test 3: PUT /posts/1

```
rutaganira@DESKTOP-BB7D3D3: /mnt/c/Users/USER$ curl -s -X PUT https://jsonplaceholder.typicode.com/posts/1 \
-H "Content-Type: application/json" \
-d '{
  "title": "Replaced Post",
  "body": "This post has been replaced completely",
  "userId": 1
}' \
-w "\nStatus Code: %{http_code}\n"
{
  "title": "Replaced Post",
  "body": "This post has been replaced completely",
  "userId": 1,
  "id": 1
}
Status Code: 200
```

Test 4: PATCH /posts/1

```

rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$ curl -s -X PATCH https://jsonplaceholder.typicode.com/posts/1 \
-H "Content-Type: application/json" \
-d '{
  "title": "Updated title"
}' \
-w "\nStatus Code: %{http_code}\n"
{
  "userId": 1,
  "id": 1,
  "title": "Updated title",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut qu
eniet architecto"
}
Status Code: 200
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$

```

Test 5: DELETE /posts/1

```

rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$ curl -s -X DELETE https://jsonplaceholder.typicode.com/posts/1 \
-w "\nStatus Code: %{http_code}\n"
{}
Status Code: 200
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$

```

Test 5: GET invalid resource /posts/99999 (simulate 404)

```

rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$ curl -s -X GET https://jsonplaceholder.typicode.com/posts/99999 \
-w "\nStatus Code: %{http_code}\n"
{}
Status Code: 404
rutaganira@DESKTOP-BB7D3DM:/mnt/c/Users/USER$

```

6. Load Balancing Simulation

Two local web servers running on different ports (**8081 and 8082**) were configured behind an Nginx load balancer.

Traffic was distributed using a round-robin method. One server was stopped to observe how traffic handling changed.

Starting server 1 and creating page for server 1

```
rutaganira@DESKTOP-BB7D3C: ~/server1$ ls
index.html
rutaganira@DESKTOP-BB7D3C: ~/server1$ cat index.html
Hello from Server 1
rutaganira@DESKTOP-BB7D3C: ~/server1$ python3 -m http.server 8081
Serving HTTP on 0.0.0.0 port 8081 (http://0.0.0.0:8081/) ...
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
```

Starting server 2 and creating page for server 2

```
rutaganira@DESKTOP-BB7D3C: ~/server2$ ls
index.html
rutaganira@DESKTOP-BB7D3C: ~/server2$ cat index.html
Hello from Server 2
rutaganira@DESKTOP-BB7D3C: ~/server2$ python3 -m http.server 8082
Serving HTTP on 0.0.0.0 port 8082 (http://0.0.0.0:8082/) ...
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:40] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:52:53] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:05] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:12] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [21/Jan/2026 23:53:27] "GET / HTTP/1.0" 200 -
```

Load balancer in action

```
rutaganira@DESKTOP-BB7D3C x rutaganira@DESKTOP-BB7D3C x rutaganira@DESKTOP-BB7D3C x
rutaganira@DESKTOP-BB7D3DM:~$ cat /etc/nginx/sites-available/default
upstream backend_pool {
    server 127.0.0.1:8081;
    server 127.0.0.1:8082;
}

server {
    listen 80 default_server;
    listen [::]:80 default_server;

    server_name _;
    location / {
        proxy_http_version 1.1;
        proxy_set_header Connection close;

        proxy_pass http://backend_pool;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
rutaganira@DESKTOP-BB7D3DM:~$ for i in {0..9}; do curl localhost; done
Hello from Server 1
Hello from Server 2
Hello from Server 1
Hello from Server 2
Hello from Server 1
Hello from Server 2
Hello from Server 1
Hello from Server 2
Hello from Server 1
Hello from Server 2
rutaganira@DESKTOP-BB7D3DM:~$ |
```

7. Security Configuration

7.1 SSL/TLS (HTTPS)

A self-signed SSL certificate was generated using OpenSSL and configured in Nginx. HTTP requests were redirected to HTTPS to enforce encrypted connections.

```
rutaganira@DESKTOP-BB7D3C x rutaganira@DESKTOP-BB7D3C x rutaganira@DESKTOP-BB7D3C x + v
rutaganira@DESKTOP-BB7D3DM:~$ curl -JLO https://dl.filippo.io/mkcert/latest?for=linux/amd64
chmod +x mkcert-v*-linux-amd64
sudo mv mkcert-v*-linux-amd64 /usr/local/bin/mkcert
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 111 100 111 0 0 105 0 0:00:01 0:00:01 --:--:-- 105
0 0 0 0 0 0 0 0 --:--:-- 0:00:01 --:--:-- 0
100 4676k 100 4676k 0 0 1246k 0 0:00:03 0:00:03 --:--:-- 3165k
rutaganira@DESKTOP-BB7D3DM:~$ mkcert -install
Created a new local CA 🌟
The local CA is now installed in the system trust store! ⚡

rutaganira@DESKTOP-BB7D3DM:~$ mkdir -p ~/certs
cd ~/certs
rutaganira@DESKTOP-BB7D3DM:~/certs$ mkcert localhost

Created a new certificate valid for the following names 📄
- "localhost"

The certificate is at "./localhost.pem" and the key at "./localhost-key.pem" ✅

It will expire on 22 April 2028 📅
rutaganira@DESKTOP-BB7D3DM:~/certs$ |
```

7.2 Firewall Rules

UFW (Uncomplicated Firewall) was configured with a default deny policy for incoming traffic. Specific rules were added to allow SSH (port 22), HTTP (port 80), and HTTPS (port 443).

```

rutaganira@DESKTOP-BB7D3DM:~$ sudo ufw allow 22/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw enable
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
rutaganira@DESKTOP-BB7D3DM:~$ sudo ufw status numbered
Status: active

    To Action From
    --
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 443/tcp ALLOW IN Anywhere
[ 4] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 5] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 6] 443/tcp (v6) ALLOW IN Anywhere (v6)

rutaganira@DESKTOP-BB7D3DM:~$

```

Access after rules were applied

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
rutaganira@DESKTOP-BB7D3DM:~$ sudo ss -lntp | grep -E ':80|443'
LISTEN 0      511          0.0.0.0:*        users:((("nginx",pid=8267,fd=7),("nginx",pid=8266,fd=7),("nginx",pid=8265,fd=7),("nginx",pid=8263,fd=7),("nginx",pid=8262,fd=7),("nginx",pid=8261,fd=7),("nginx",pid=8259,fd=7),("nginx",pid=8258,fd=7),("nginx",pid=8256,fd=7),("nginx",pid=8255,fd=7),("nginx",pid=8254,fd=7))
LISTEN 0      511          0.0.0.0:*        users:((("nginx",pid=8267,fd=5),("nginx",pid=8266,fd=5),("nginx",pid=8265,fd=5),("nginx",pid=8263,fd=5),("nginx",pid=8262,fd=5),("nginx",pid=8261,fd=5),("nginx",pid=8260,fd=5),("nginx",pid=8259,fd=5),("nginx",pid=8258,fd=5),("nginx",pid=8256,fd=5),("nginx",pid=8255,fd=5),("nginx",pid=8254,fd=5))
LISTEN 0      511          [::]:443         [::]:* users:((("nginx",pid=8267,fd=8),("nginx",pid=8266,fd=8),("nginx",pid=8265,fd=8),("nginx",pid=8263,fd=8),("nginx",pid=8262,fd=8),("nginx",pid=8261,fd=8),("nginx",pid=8260,fd=8),("nginx",pid=8259,fd=8),("nginx",pid=8258,fd=8),("nginx",pid=8257,fd=8),("nginx",pid=8256,fd=8),("nginx",pid=8255,fd=8),("nginx",pid=8254,fd=8))
LISTEN 0      511          [::]:80          [::]:* users:((("nginx",pid=8267,fd=6),("nginx",pid=8266,fd=6),("nginx",pid=8265,fd=6),("nginx",pid=8263,fd=6),("nginx",pid=8262,fd=6),("nginx",pid=8261,fd=6),("nginx",pid=8260,fd=6),("nginx",pid=8259,fd=6),("nginx",pid=8258,fd=6),("nginx",pid=8257,fd=6),("nginx",pid=8256,fd=6),("nginx",pid=8255,fd=6),("nginx",pid=8254,fd=6))
rutaganira@DESKTOP-BB7D3DM:~$ openssl s_client -connect localhost:443 -servername localhost
CONNECTED(00000003)
depth1:0 = mkcert development CA, OU = rutaganira@DESKTOP-BB7D3DM, CN = mkcert rutaganira@DESKTOP-BB7D3DM
verify return:1
depth:0 = mkcert development certificate, OU = rutaganira@DESKTOP-BB7D3DM
verify return:1
-----
Certificate chain
 0 s:O = mkcert development certificate, OU = rutaganira@DESKTOP-BB7D3DM
  i:O = mkcert development CA, OU = rutaganira@DESKTOP-BB7D3DM, CN = mkcert rutaganira@DESKTOP-BB7D3DM
   a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
   v:NotBefore: Jan 21 20:18:56 2026 GMT; NotAfter: Apr 21 20:18:56 2028 GMT
-----
Server certificate
-----BEGIN CERTIFICATE-----
MIIEMDCCApCgAwIBAgIRAKiGIFj+9STkiLr/q0YvZYwDQVJKoZIhvcNAQELBQAw
cTEeMBwwGAQGMVbWwtjZXJ0IGRldmVsb3ZzBTU5ODIENBMwIQYDVQQDL2pydXRh
Z2FuaXJ0QERFU0tUT1AtQkI3RDNETTETMcGCGA1UEAhmbWwtjZXJ0IHJlZGFyVS5p
cmMFAREVTj1RPUC1CQjdEM0RMRXBMTDIzMDEYMTIwMTg1Ln0XTDI4MDQyMTIwMTg1
NlowTjEnMCUGA1UEChMbWwtjZXJ0IGRldmVsb3ZzBTU5ODIENlcncRpZmljYXRlMSMw
IQYDVQQDL2pydXRhZ2FuaXJ0QERFU0tUT1AtQkI3RDNETTCASImQYDQVJKoZIhvcNAQ
AQEBBgQdggEPADCCAQCqGBANUzH2XuuoYB900iCKrf+tFP4d80aeZAozSeS03vf
QUTEXPh66kgNUHWQYHQcyd42hUXAw8QdLDDBPLbZPHMLLEyrrTNeN9mw/DpLT
10e08mKfUw0Pgl0m0TS6f0oe0qU192ayRw/Rg0EEILfmZmg0K
3wZnTLvYCYGZ2fU0ROrp3dptKLNIAnHoKA223w3qj0IDTrFIcn03eaUsICB3arpBH
JmZDe H4jn0Da57vsrNl(qoN5UjvshMtRoapooMdab5fRl/pQnQT13B6gaCysem0Z
```


8. Debugging Scenarios

Three common infrastructure problems were simulated and resolved:

1. Port conflict

HA proxy was occupying the required port(80), preventing Nginx from starting.

Resolved by identifying and stopping the conflicting process.

Ports conflict

```
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Thu 2026-01-22 21:42:44 +04; 9s ago
 Duration: 6min 51.029s
    Docs: man:nginx(8)
 Process: 1576 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Process: 1578 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=1/FAILURE)
    CPU: 28ms

Jan 22 21:42:41 DESKTOP-BB7D3DM systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 22 21:42:42 DESKTOP-BB7D3DM nginx[1578]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Jan 22 21:42:42 DESKTOP-BB7D3DM nginx[1578]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Jan 22 21:42:43 DESKTOP-BB7D3DM nginx[1578]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Jan 22 21:42:43 DESKTOP-BB7D3DM nginx[1578]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Jan 22 21:42:44 DESKTOP-BB7D3DM nginx[1578]: nginx: [emerg] still could not bind()
Jan 22 21:42:44 DESKTOP-BB7D3DM systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Jan 22 21:42:44 DESKTOP-BB7D3DM systemd[1]: nginx.service: Failed with result 'exit-code'.
Jan 22 21:42:44 DESKTOP-BB7D3DM systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.
```

Stopped conflicting process

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo ss -ltnp | grep -E :80
LISTEN 0      4096      0.0.0.0:80      0.0.0.0:*      users:((("haproxy",pid=1556,fd=10))
rutaganira@DESKTOP-BB7D3DM:~$ sudo service haproxy stop
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx restart
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx status
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-22 21:44:40 +04; 15s ago
    Docs: man:nginx(8)
 Process: 1612 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Process: 1613 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Main PID: 1615 (nginx)
      Tasks: 13 (limit: 9287)
   Memory: 9.4M (peak: 11.1M)
      CPU: 57ms
   CGroup: /system.slice/nginx.service
           └─1615 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1616 "nginx: worker process"
               └─1617 "nginx: worker process"
                 └─1618 "nginx: worker process"
                   └─1619 "nginx: worker process"
                     └─1620 "nginx: worker process"
                       └─1621 "nginx: worker process"
                         └─1622 "nginx: worker process"
                           └─1623 "nginx: worker process"
                             └─1624 "nginx: worker process"
                               └─1625 "nginx: worker process"
                                 └─1627 "nginx: worker process"
                                   └─1628 "nginx: worker process"

Jan 22 21:44:40 DESKTOP-BB7D3DM systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 22 21:44:40 DESKTOP-BB7D3DM systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

2. File missing values

We deleted a curly bracket in the nginx configuration file which cause an error in restarting the server.

Nginx configuration file missing expected values

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo vi /etc/nginx/sites-available/default
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx restart
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
rutaganira@DESKTOP-BB7D3DM:~$ systemctl status nginx.service
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Thu 2026-01-22 21:32:14 +04; 8s ago
  Duration: 7min 34.300s
    Docs: man:nginx(8)
   Process: 1010 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=1/FAILURE)
      CPU: 14ms

Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: nginx.service: Deactivated successfully.
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 22 21:32:14 DESKTOP-BB7D3DM nginx[1010]: 2026/01/22 21:32:14 [emerg] 1010#1010: unexpected end of file, expecting "]" in /etc/nginx/sites-enabled/default
Jan 22 21:32:14 DESKTOP-BB7D3DM nginx[1010]: nginx: configuration file /etc/nginx/nginx.conf test failed
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: nginx.service: Failed with result 'exit-code'.
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.
lines 1-16/16 (END)
```

Fixed by adding missing values

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo vi /etc/nginx/sites-available/default
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx restart
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
rutaganira@DESKTOP-BB7D3DM:~$ systemctl status nginx.service
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Thu 2026-01-22 21:32:14 +04; 8s ago
  Duration: 7min 34.300s
    Docs: man:nginx(8)
   Process: 1010 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=1/FAILURE)
      CPU: 14ms

Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: nginx.service: Deactivated successfully.
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 22 21:32:14 DESKTOP-BB7D3DM nginx[1010]: 2026/01/22 21:32:14 [emerg] 1010#1010: unexpected end of file, expecting "]" in /etc/nginx/sites-enabled/default
Jan 22 21:32:14 DESKTOP-BB7D3DM nginx[1010]: nginx: configuration file /etc/nginx/nginx.conf test failed
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: nginx.service: Failed with result 'exit-code'.
Jan 22 21:32:14 DESKTOP-BB7D3DM systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.

rutaganira@DESKTOP-BB7D3DM:~$ sudo vi /etc/nginx/sites-available/default
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx restart
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx restart
rutaganira@DESKTOP-BB7D3DM:~$ sudo service nginx reload
rutaganira@DESKTOP-BB7D3DM:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
rutaganira@DESKTOP-BB7D3DM:~$
```

3. Conflict port disabled

We disabled port 8081 where one of our servers were connected and when trying to connect again the connection refused. We again added the allow rule to the port hence allowing connection again.

Connection to port 8081 disabled

```
rutaganira@DESKTOP-BB7D3DM:~$ curl localhost:8081
curl: (7) Failed to connect to localhost port 8081 after 0 ms: Couldn't connect to server
rutaganira@DESKTOP-BB7D3DM:~$
```

Allowing port 8081

```
rutaganira@DESKTOP-BB7D3DM:~$ sudo ufw allow 8081/tcp
Rules updated
Rules updated (v6)
rutaganira@DESKTOP-BB7D3DM:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
rutaganira@DESKTOP-BB7D3DM:~$ curl localhost:8081
Hello from Server 1
rutaganira@DESKTOP-BB7D3DM:~$
```

9. Conclusion

This project provided us with hands-on experience in configuring and securing a web infrastructure. The tasks covered essential production environment skills including secure remote access, web server management, traffic distribution, and systematic troubleshooting. Which helped us form the foundation of real-world IT infrastructure operations.